

GHK-Cu Topical Serum

Once or Twice Daily on Face & Scalp

GHK-Cu is widely used in cosmetic formulations to improve several measures of photoaged facial skin and support hair growth and thickness.

- ★★ Effective for improving several measures of photoaged facial skin.
- ★★ Effective for increasing hair growth and thickness.
- ★★ Effective for supporting healing of skin ulcers and wound healing.
- ★★ Short-term topical use is generally well tolerated, long-term topical safety is plausible.

Disclaimer

This document is an evidence-scoped research survey, not medical advice or a substitute for individualized care. It distinguishes human, animal, in vitro, mechanistic, and community evidence so that each claim is stated at the strongest level the underlying evidence supports. Topical cosmetic use should be separated from treatment of wounds or medical skin conditions, which requires an appropriate clinical setting.

Human testing is often limited by formulation differences, commercial ownership, cost, and the difficulty of measuring long-term cosmetic outcomes. The absence of a human trial for a specific formulation does not by itself invalidate a biological effect, but it limits how confidently that effect can be generalized.

This document was updated 5-Sep-26.

What GHK Cu Is

GHK-Cu is the copper complex of glycyl-L-histidyl-L-lysine, a naturally occurring human tripeptide. It is not a classical receptor agonist. By binding and transporting copper and interacting with extracellular-matrix, inflammatory, antioxidant, and repair pathways, it can influence multiple signaling processes involved in skin maintenance and wound repair.

Formulation

- Finished formulation: 2% w/w GHK-Cu raw powder in an aqueous cosmetic base containing 2% total mixed high- and low-molecular-weight hyaluronic acid.
- A 2% w/w GHK-Cu concentration equals 20 mg of raw powder per gram of finished serum. The amount of active GHK-Cu depends on the raw material's assay, counterion or salt form, water content, and certificate of analysis.
- The 2% hyaluronic-acid value refers to total hyaluronic acid in the finished base. The high- to low-molecular-weight ratio depends on the supplier's specification.

Use Patterns

- Twice-daily application for 12 weeks is the best-represented pattern in the available facial and eyebrow studies. Once-daily application is a practical lower-frequency pattern, but comparative evidence is unavailable.
- Apply a thin, even layer to clean skin. Avoid direct contact with the eyes and mucosal surfaces.
- Patch testing on a small area is appropriate before broader facial use, particularly for reactive skin or a newly prepared batch.

Benefits

- **Photoaged skin appearance:** Topical GHK-Cu improves the overall appearance of photoaged facial and periocular skin, including clarity and visible texture, in human studies using GHK-Cu creams.
- **Fine lines and wrinkles:** Topical GHK-Cu reduces facial and periocular lines and wrinkles over approximately 12 weeks in the reported human studies.

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- **Skin firmness and density:** Topical GHK-Cu improves measured skin density, thickness, firmness, and laxity in the reported facial studies.
- **Uneven pigmentation appearance:** Topical GHK-Cu improves mottled pigmentation and visible photodamage in reported human facial studies.
- **Eyebrow hair growth:** A 2% GHK-Cu serum used twice daily for 12 weeks increased eyebrow hair count and diameter and improved photographic assessments in a small randomized split-face trial.
- **Local wound repair:** A 2% GHK-Cu gel supported closure of diabetic lower-extremity ulcers when used with debridement and structured wound care. This evidence applies to the studied medical formulation and setting.
- **Collagen synthesis support:** GHK-Cu likely promotes local collagen synthesis and extracellular-matrix repair based on injected and topical animal evidence, supported by fibroblast and ex vivo skin findings.
- **Matrix remodeling support:** GHK-Cu likely supports regulated remodeling of collagen, glycosaminoglycans, and other matrix components during local repair.
- **Scar appearance support:** Improvement in scar texture and appearance is plausible through effects on collagen production, matrix remodeling, inflammation, and wound repair.
- **Hydration and barrier support:** The 2% hyaluronic-acid base hydrates the stratum corneum and supports surface barrier function. Combining it with GHK-Cu plausibly adds complementary hydration and repair effects.
- **Basement membrane support:** GHK-Cu combined with hyaluronic acid plausibly supports the dermal-epidermal junction by increasing collagen IV, based on fibroblast and ex vivo skin research.
- **Inflammatory signal regulation:** Local modulation of excessive inflammatory signaling is plausible through effects on NF- κ B-related and cytokine pathways described in preclinical research.
- **Antioxidant defense support:** Local support of antioxidant defenses is plausible through copper-dependent enzymes and GHK-Cu-associated regulation of oxidative-stress pathways.

Cautions

- Copper allergy, contact sensitivity, or sensitivity to another ingredient in the finished base warrants caution and patch testing.
- Active dermatitis, a disrupted skin barrier, or recent cosmetic procedures can increase irritation and absorption. Introduce the serum only after the skin has recovered or with clinician guidance.
- Use on chronic ulcers, burns, infected wounds, or other significant injuries belongs in supervised medical wound care rather than a cosmetic protocol.
- Application directly over a known or suspected skin cancer warrants clinical guidance because GHK-Cu participates in repair, angiogenic, and matrix-remodeling pathways.
- Pregnancy and breastfeeding use remains speculative and warrants individualized clinical guidance.
- Wilson disease or another disorder of copper handling warrants clinician guidance before prolonged or large-area use.

Cautions

- Temporary redness, stinging, itching, dryness, or irritation can occur with topical products, particularly on reactive or recently treated skin.
- Persistent rash, swelling, blistering, or worsening irritation warrants discontinuation and clinical evaluation.

Biological Mechanism

GHK binds copper with high affinity and forms GHK-Cu, a redox-active but biologically regulated complex. In skin models, GHK-Cu influences fibroblast and keratinocyte activity, collagen and glycosaminoglycan production, matrix metalloproteinases and their inhibitors, inflammatory cytokine signaling, antioxidant

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defenses, angiogenesis, and recruitment of repair processes. These overlapping effects support regulated tissue remodeling rather than a single linear pathway.

Topical activity depends on the amount of intact complex that reaches viable epidermal and dermal compartments. GHK-Cu is hydrophilic, and passive passage across intact stratum corneum can be limited. Vehicle composition, concentration, exposure time, skin condition, and delivery technology therefore influence whether mechanistic activity translates into a visible clinical effect.

Formulation Chemistry and Hyaluronic Acid

The proposed serum combines 2% GHK-Cu with 2% total mixed-molecular-weight hyaluronic acid in an aqueous cosmetic base. Hyaluronic acid is a water-binding polymer that increases surface hydration and can improve slip and film formation. Higher-molecular-weight fractions primarily support surface hydration and film properties, while lower-molecular-weight fractions interact more readily with superficial skin compartments. Their functional behavior depends on molecular-weight distribution, concentration, and formulation.

In fibroblast and ex vivo skin experiments, GHK-Cu combined with hyaluronic acid increased collagen I, IV, and VII expression, with the strongest collagen IV response reported for a 1:9 GHK-Cu to low-molecular-weight hyaluronic-acid ratio. This provides a plausible biological rationale for combining the ingredients, but it does not establish that every mixed-molecular-weight 2% HA base reproduces that ratio or delivery profile.

GHK-Cu stability is sensitive to formulation conditions. Published preformulation and ex vivo work supports a mildly acidic finished pH, and one accelerated-aging experiment found less than 5% degradation at pH 5.5 to 6.5. The finished serum should therefore be assessed as a complete product, including pH, preservative compatibility, raw-material assay, color and odor changes, container compatibility, and microbial stability.

A concentration written as 2% should specify weight by weight. For a finished batch, 2% w/w means 2 g of raw powder per 100 g of serum. Converting this value to milligrams per milliliter requires the measured density of the finished product.

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